

Basic Structures: Sets, Functions, Sequences, Sums, and Matrices

Chapter 2

Chapter Summary

- Sets
 - The Language of Sets
 - Set Operations
 - Set Identities
- Functions
 - Types of Functions
 - Operations on Functions
- Sequences and Summations
 - Types of Sequences
 - Summation Formulae
- Set Cardinality
- Matrices
 - Matrix Arithmetic

Sets

- Sets are one of the basic building blocks for the types of objects considered in discrete mathematics.
 - Important for counting.
 - Programming languages have set operations.
- Set theory is an important branch of mathematics.
 - Many different systems of axioms have been used to develop set theory.
 - Here we are not concerned with a formal set of axioms for set theory. Instead, we will use what is called naïve set theory.

Sets

- A *set* is an unordered collection of objects.
 - the students in this class
 - the chairs in this room
- The objects in a set are called the *elements*, or *members* of the set. A set is said to *contain* its elements.
- The notation $a \in A$ denotes that a is an element of the set A .
- If a is not a member of A , write $a \notin A$

Describing a Set: Roster Method

- $S = \{a, b, c, d\}$
- Order not important

$$S = \{a, b, c, d\} = \{b, c, a, d\}$$

- Each distinct object is either a member or not; listing more than once does not change the set.

$$S = \{a, b, c, d\} = \{a, b, c, b, c, d\}$$

- Ellipses (...) may be used to describe a set without listing all of the members when the pattern is clear.

$$S = \{a, b, c, d, \dots, z\}$$

Roster Method

- Set of all vowels in the English alphabet: $V = \{a, e, i, o, u\}$
- Set of all odd positive integers less than 10: $O = \{1, 3, 5, 7, 9\}$
- Set of all positive integers less than 100: $S = \{1, 2, 3, \dots, 99\}$
- Set of all integers less than 0: $S = \{\dots, -3, -2, -1\}$

Some Important Sets

$\mathbf{N}$ = *natural numbers* = $\{0, 1, 2, 3, \dots\}$

$\mathbf{Z}$ = *integers* = $\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$

$\mathbf{Z}^+$ = *positive integers* = $\{1, 2, 3, \dots\}$

$\mathbf{R}$ = *set of real numbers*

$\mathbf{R}^+$ = *set of positive real numbers*

$\mathbf{C}$ = *set of complex numbers.*

$\mathbf{Q}$ = *set of rational numbers*

Set-Builder Notation

- Specify the property or properties that all members must satisfy:

$$S = \{x \mid x \text{ is a positive integer less than } 100\}$$

$$O = \{x \mid x \text{ is an odd positive integer less than } 10\}$$

$$O = \{x \in \mathbf{Z}^+ \mid x \text{ is odd and } x < 10\}$$

- A predicate may be used:

$$S = \{x \mid P(x)\}$$

- Example: $S = \{x \mid \text{Prime}(x)\}$

- Positive rational numbers:

$$\mathbf{Q}^+ = \{x \in \mathbf{R} \mid x = p/q, \text{ for some positive integers } p, q\}$$

Interval Notation

$$[a, b] = \{x \mid a \leq x \leq b\}$$

$$[a, b) = \{x \mid a \leq x < b\}$$

$$(a, b] = \{x \mid a < x \leq b\}$$

$$(a, b) = \{x \mid a < x < b\}$$

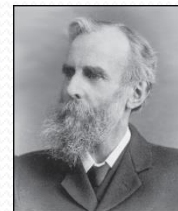
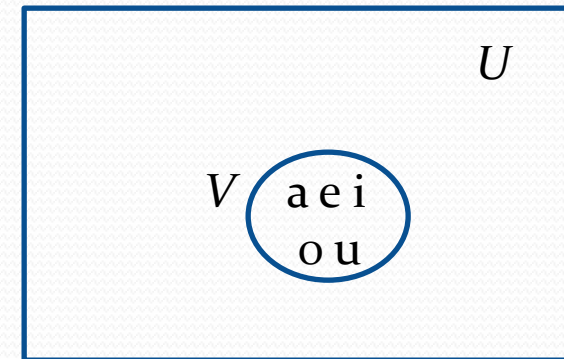
closed interval $[a, b]$,

open interval (a, b)

Universal Set and Empty Set

- The *universal set* U is the set containing everything currently under consideration.
 - Sometimes implicit
 - Sometimes explicitly stated.
 - Contents depend on the context.
- The empty set is the set with no elements. Symbolized $\emptyset$, but $\{\}$ also used.

Venn Diagram



John Venn (1834-1923)
Cambridge, UK

Some things to remember

- Sets can be elements of sets.

$$\{\{1,2,3\}, a, \{b,c\}\}$$

$$\{\mathbf{N}, \mathbf{Z}, \mathbf{Q}, \mathbf{R}\}$$

- The empty set is different from a set containing the empty set.

$$\emptyset \neq \{ \emptyset \}$$

Set Equality

Definition: Two sets are *equal* if and only if they have the same elements.

- Therefore if A and B are sets, then A and B are equal if and only if $\forall x (x \in A \leftrightarrow x \in B)$
- We write $A = B$ if A and B are equal sets.

$$\{1, 3, 5\} = \{3, 5, 1\}$$

$$\{1, 5, 5, 5, 3, 3, 1\} = \{1, 3, 5\}$$

Subsets

Definition: The set A is a *subset* of B , if and only if every element of A is also an element of B .

- The notation $A \subseteq B$ is used to indicate that A is a subset of the set B .
- $A \subseteq B$ holds if and only if $\forall x(x \in A \rightarrow x \in B)$ is true.
 1. Because $a \in \emptyset$ is always false, $\emptyset \subseteq S$, for every set S .
 2. Because $a \in S \rightarrow a \in S$, $S \subseteq S$, for every set S .

Another look at Equality of Sets

- Recall that two sets A and B are *equal*, denoted by $A = B$, iff
$$\forall x (x \in A \leftrightarrow x \in B)$$

- Using logical equivalences we have that $A = B$ iff

$$\forall x [(x \in A \rightarrow x \in B) \wedge (x \in B \rightarrow x \in A)]$$

- This is equivalent to

$$A \subseteq B \quad \text{and} \quad B \subseteq A$$

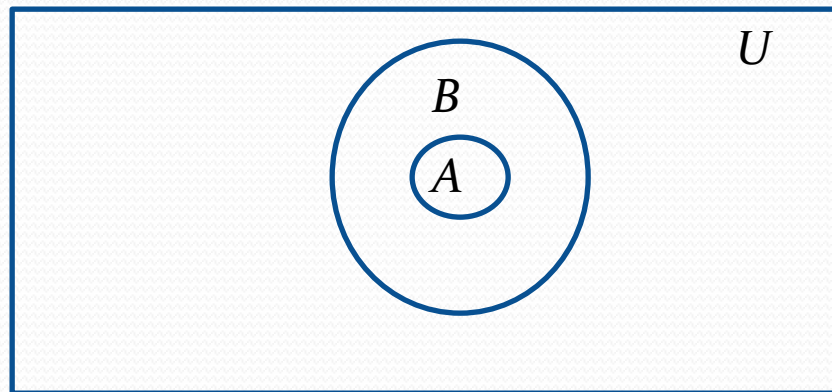
Proper Subsets

Definition: If $A \subseteq B$, but $A \neq B$, then we say A is a *proper subset* of B , denoted by $A \subset B$. If $A \subset B$, then

$$\forall x(x \in A \rightarrow x \in B) \wedge \exists x(x \in B \wedge x \notin A)$$

is true.

Venn Diagram



Set Cardinality

Definition: If there are exactly n distinct elements in S where n is a nonnegative integer, we say that S is *finite*. Otherwise it is *infinite*.

Definition: The *cardinality* of a finite set A , denoted by $|A|$, is the number of (distinct) elements of A .

Examples:

1. $|\emptyset| = 0$
2. Let S be the letters of the English alphabet. Then $|S| = 26$
3. $|\{1,2,3\}| = 3$
4. $|\{\emptyset\}| = 1$
5. The set of integers is infinite.

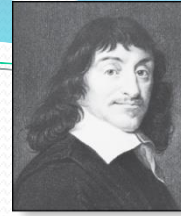
Power Sets

Definition: The set of all subsets of a set A , denoted $\mathcal{P}(A)$, is called the *power set* of A .

Example: If $A = \{a, b\}$ then

$$\mathcal{P}(A) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$$

- If a set has n elements, then the cardinality of the power set is 2^n . (In Chapters 5 and 6, we will discuss different ways to show this.)



René Descartes
(1596-1650)

Cartesian Product

Definition: The *Cartesian Product* of two sets A and B , denoted by $A \times B$ is the set of ordered pairs (a,b) where $a \in A$ and $b \in B$.

$$A \times B = \{(a, b) | a \in A \wedge b \in B\}$$

Example:

$$A = \{a, b\} \quad B = \{1, 2, 3\}$$

$$A \times B = \{(a, 1), (a, 2), (a, 3), (b, 1), (b, 2), (b, 3)\}$$

- **Definition:** A subset R of the Cartesian product $A \times B$ is called a *relation* from the set A to the set B . (Relations will be covered in depth in Chapter 9.)

Union

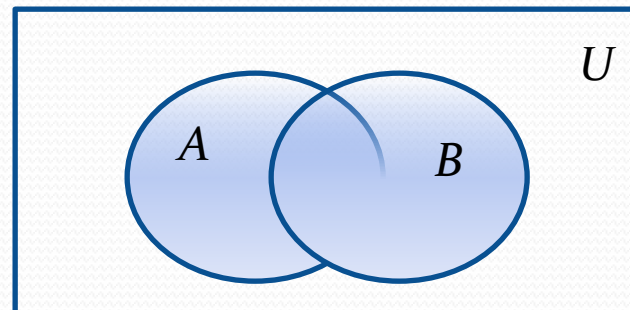
- **Definition:** Let A and B be sets. The *union* of the sets A and B , denoted by $A \cup B$, is the set:

$$\{x \mid x \in A \vee x \in B\}$$

- **Example:** What is $\{1,2,3\} \cup \{3,4,5\}$?

Solution: $\{1,2,3,4,5\}$

Venn Diagram for $A \cup B$



Intersection

- **Definition:** The *intersection* of sets A and B , denoted by $A \cap B$, is

$$\{x | x \in A \wedge x \in B\}$$

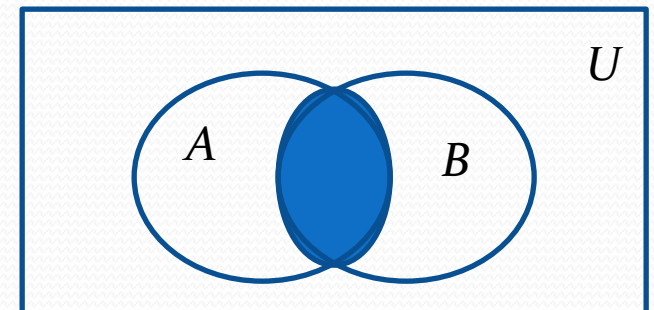
- Note if the intersection is empty, then A and B are said to be *disjoint*.
- **Example:** What is? $\{1,2,3\} \cap \{3,4,5\}$?

Solution: $\{3\}$

- **Example:** What is?
 $\{1,2,3\} \cap \{4,5,6\}$?

Solution: $\emptyset$

Venn Diagram for $A \cap B$



Complement

Definition: If A is a set, then the complement of the A (with respect to U), denoted by $\bar{A}$ is the set $U - A$

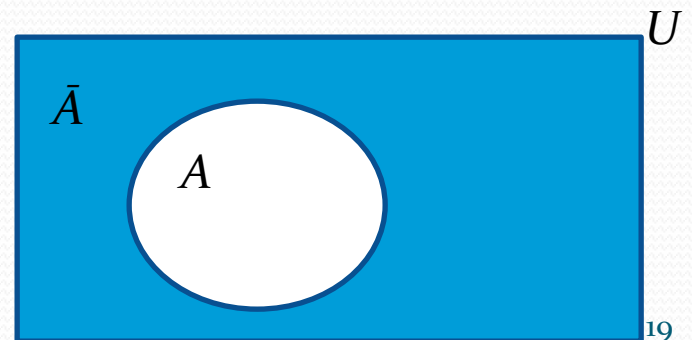
$$\bar{A} = \{x \in U \mid x \notin A\}$$

(The complement of A is sometimes denoted by A^c .)

Example: If U is the positive integers less than 100, what is the complement of $\{x \mid x > 70\}$

Solution: $\{x \mid x \leq 70\}$

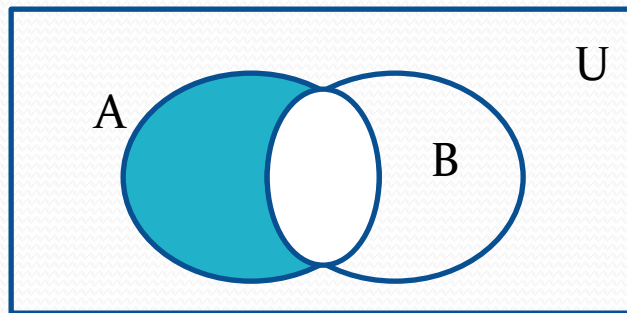
Venn Diagram for Complement



Difference

- **Definition:** Let A and B be sets. The *difference* of A and B , denoted by $A - B$, is the set containing the elements of A that are not in B . The difference of A and B is also called the complement of B with respect to A .

$$A - B = \{x \mid x \in A \wedge x \notin B\} = A \cap \bar{B}$$

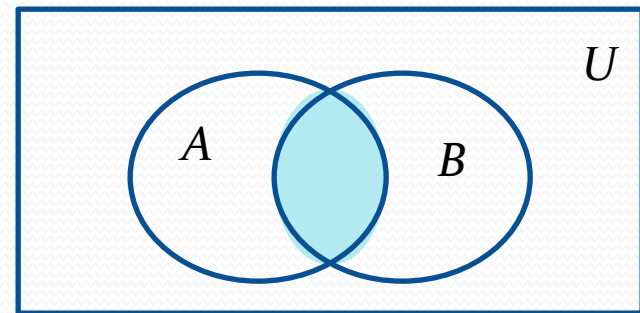


Venn Diagram for $A - B$

The Cardinality of the Union of Two Sets

- Inclusion-Exclusion

$$|A \cup B| = |A| + |B| - |A \cap B|$$



Venn Diagram for A , B , $A \cap B$, $A \cup B$

- **Example:** Let A be the math majors in your class and B be the CS majors. To count the number of students who are either math majors or CS majors, add the number of math majors and the number of CS majors, and subtract the number of joint CS/math majors.
- We will return to this principle in Chapter 6 and Chapter 8 where we will derive a formula for the cardinality of the union of n sets, where n is a positive integer.

Review Questions

Example: $U = \{0,1,2,3,4,5,6,7,8,9,10\}$ $A = \{1,2,3,4,5\}$, $B = \{4,5,6,7,8\}$

1. $A \cup B$

Solution: $\{1,2,3,4,5,6,7,8\}$

2. $A \cap B$

Solution: $\{4,5\}$

3. $\bar{A}$

Solution: $\{0,6,7,8,9,10\}$

4. $\bar{B}$

Solution: $\{0,1,2,3,9,10\}$

5. $A - B$

Solution: $\{1,2,3\}$

6. $B - A$

Solution: $\{6,7,8\}$

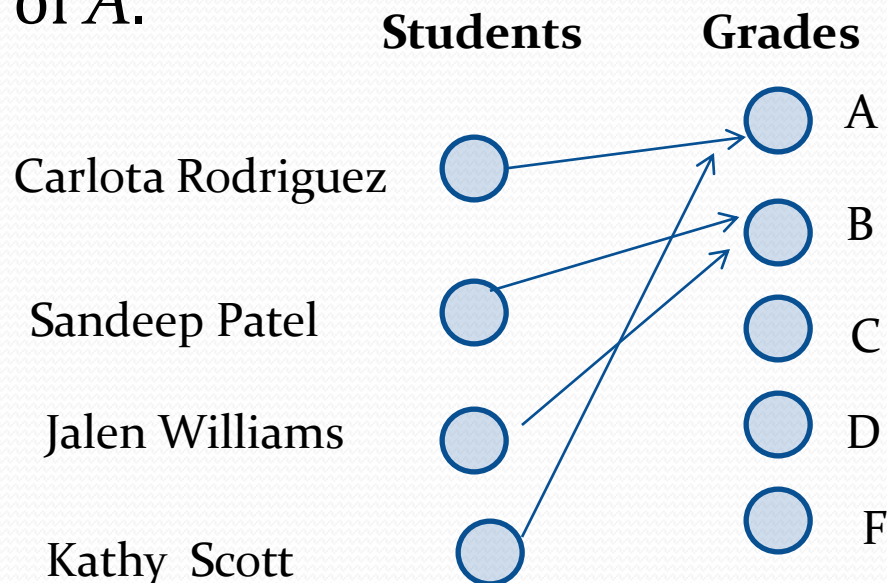
Set Identities

<i>name</i>	<i>rule</i>
<i>Commutative laws</i>	$A \cap B = B \cap A \quad A \cup B = B \cup A$
<i>Associative laws</i>	$A \cap (B \cap C) = (A \cap B) \cap C$ $A \cup (B \cup C) = (A \cup B) \cup C$
<i>Distributive laws</i>	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
<i>DeMorgan's laws</i>	$\overline{A \cap B} = \overline{A} \cup \overline{B} \quad \overline{A \cup B} = \overline{A} \cap \overline{B}$
<i>Complement laws</i>	$A \cap \overline{A} = \emptyset \quad A \cup \overline{A} = U$
<i>Double complement law</i>	$\overline{\overline{A}} = A$
<i>Idempotent laws</i>	$A \cap A = A \quad A \cup A = A$
<i>Absorption laws</i>	$A \cap (A \cup B) = A \quad A \cup (A \cap B) = A$
<i>Dominance laws</i>	$A \cap \emptyset = \emptyset \quad A \cup U = U$
<i>Identity laws</i>	$A \cup \emptyset = A \quad A \cap U = A$

Functions

Definition: Let A and B be nonempty sets. A *function* f from A to B , denoted $f: A \rightarrow B$ is an assignment of each element of A to exactly one element of B . We write $f(a) = b$ if b is the unique element of B assigned by the function f to the element a of A .

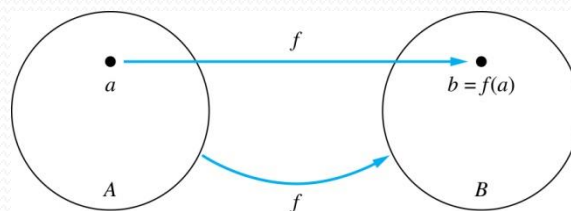
- Functions are sometimes called *mappings* or *transformations*.



Functions

Given a function $f: A \rightarrow B$:

- We say f maps A to B or f is a *mapping* from A to B .
- A is called the *domain* of f .
- B is called the *codomain* of f .
- If $f(a) = b$,
 - then b is called the *image* of a under f .
 - a is called the *preimage* of b .
- The range of f is the set of all images of points in A under f . We denote it by $f(A)$.
- Two functions are *equal* when they have the same domain, the same codomain and map each element of the domain to the same element of the codomain.



Questions

$f(a) = ?$ z

The image of d is ? z

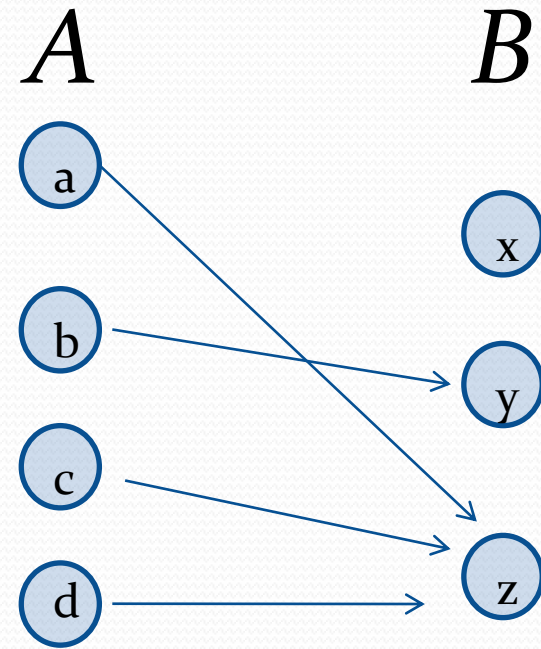
The domain of f is ? A

The codomain of f is ? B

The preimage of y is ? b

$f(A) = ?$ $\{y, z\}$

The preimage(s) of z is (are) ? $\{a, c, d\}$



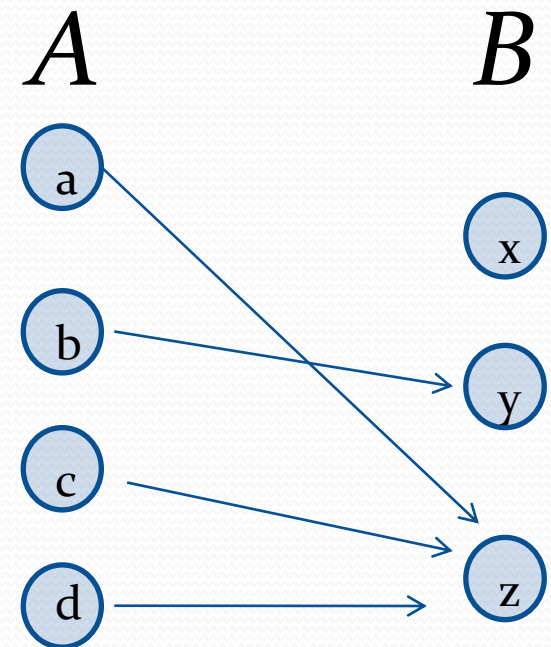
Question on Functions and Sets

- If $f : A \rightarrow B$ and S is a subset of A , then

$$f(S) = \{f(s) | s \in S\}$$

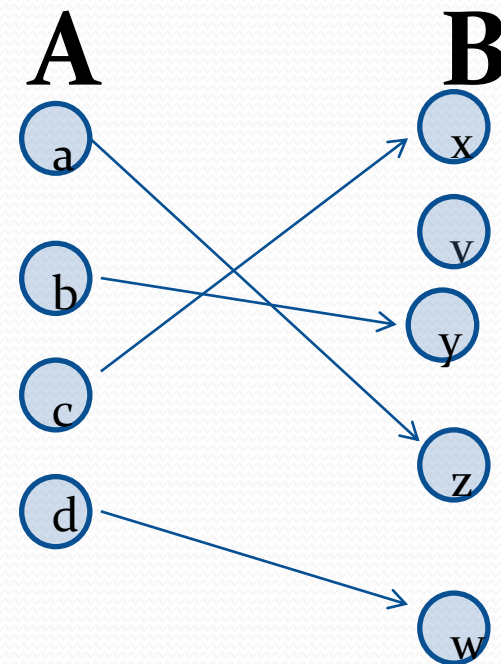
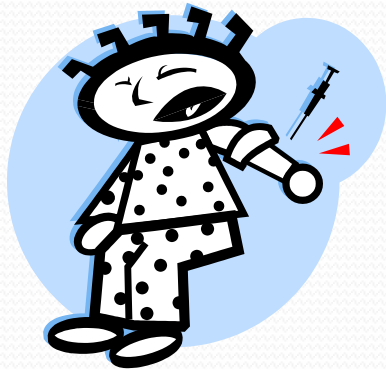
$f\{a,b,c\}$ is ? $\{y,z\}$

$f\{c,d\}$ is ? $\{z\}$



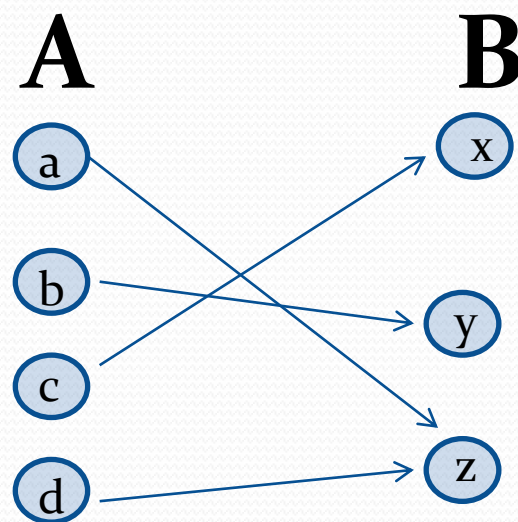
Injectons

Definition: A function f is said to be *one-to-one*, or *injective*, if and only if $f(a) = f(b)$ implies that $a = b$ for all a and b in the domain of f . A function is said to be an *injection* if it is one-to-one.



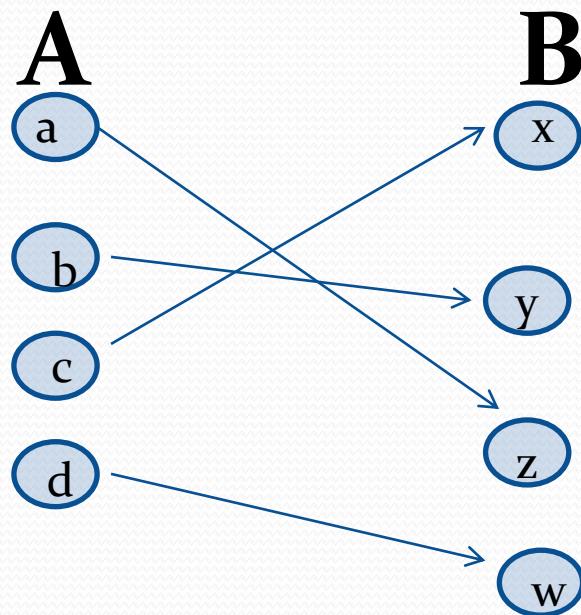
Surjections

Definition: A function f from A to B is called *onto* or *surjective*, if and only if for every element $b \in B$ there is an element $a \in A$ with $f(a) = b$. A function f is called a *surjection* if it is *onto*.



Bijections

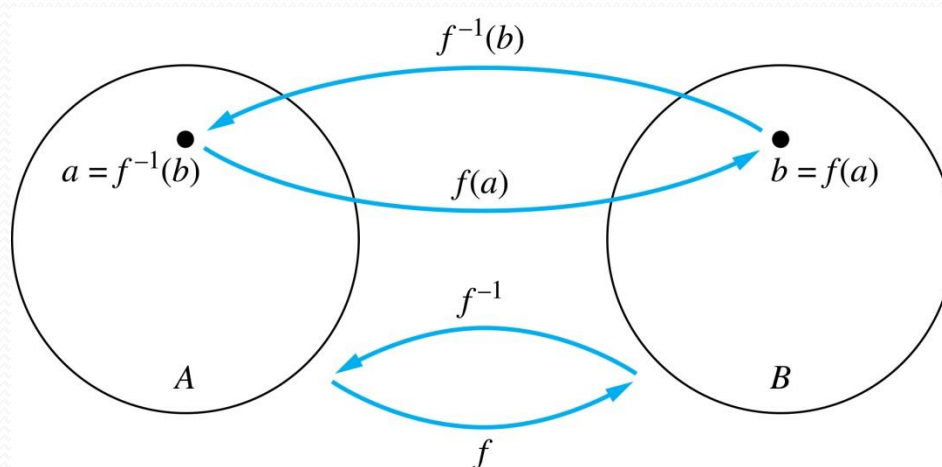
Definition: A function f is a *one-to-one correspondence*, or a *bijection*, if it is both one-to-one and onto (surjective and injective).



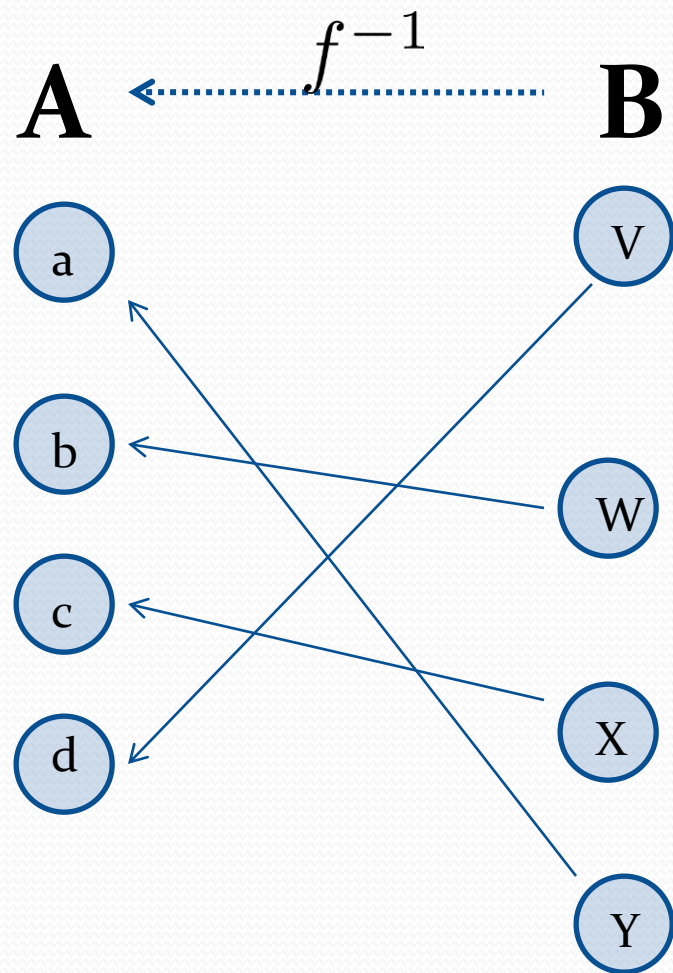
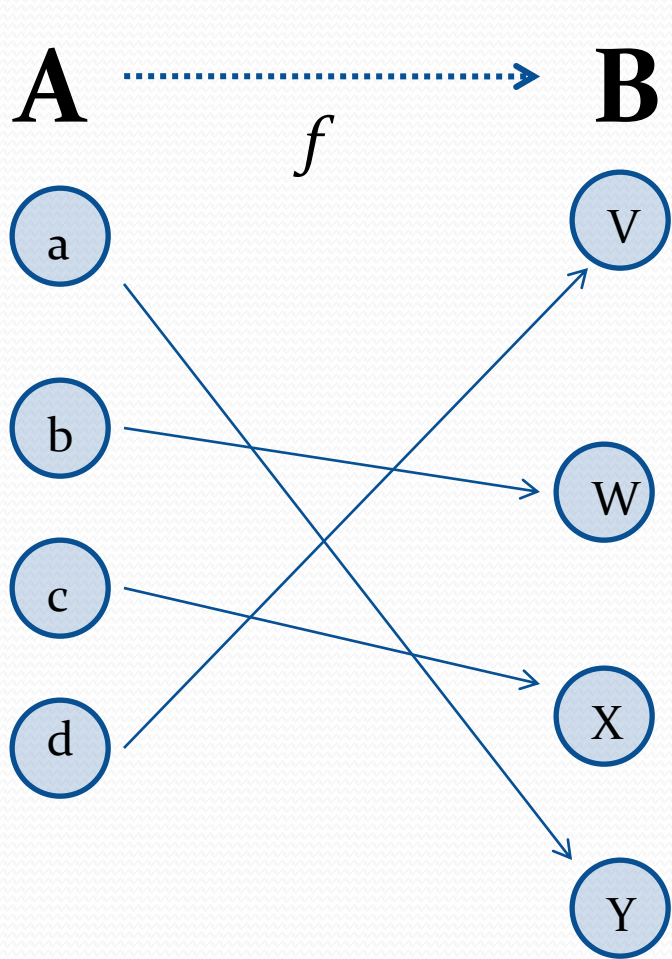
Inverse Functions

Definition: Let f be a bijection from A to B . Then the *inverse* of f , denoted f^{-1} , is the function from B to A defined as $f^{-1}(y) = x$ iff $f(x) = y$

No inverse exists unless f is a bijection. Why?



Inverse Functions



Questions

Example 1: Let f be the function from $\{a, b, c\}$ to $\{1, 2, 3\}$ such that $f(a) = 2$, $f(b) = 3$, and $f(c) = 1$. Is f invertible and if so what is its inverse?

Solution: The function f is invertible because it is a one-to-one correspondence. The inverse function f^{-1} reverses the correspondence given by f , so $f^{-1}(1) = c$, $f^{-1}(2) = a$, and $f^{-1}(3) = b$.

Questions

Example 2: Let $f: \mathbf{Z} \rightarrow \mathbf{Z}$ be such that $f(x) = x + 1$. Is f invertible, and if so, what is its inverse?

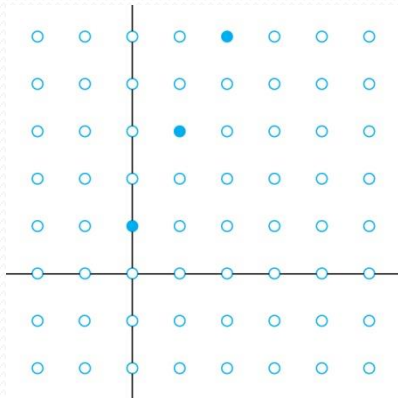
Solution: The function f is invertible because it is a one-to-one correspondence. The inverse function f^{-1} reverses the correspondence so $f^{-1}(y) = y - 1$.

Example 3: Let $f: \mathbf{R} \rightarrow \mathbf{R}$ be such that $f(x) = x^2$. Is f invertible, and if so, what is its inverse?

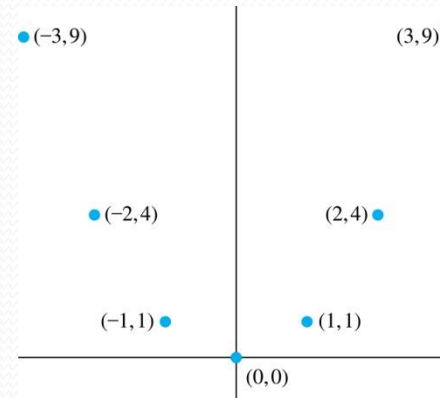
Solution: The function f is not invertible because it is not one-to-one .

Graphs of Functions

- Let f be a function from the set A to the set B . The *graph* of the function f is the set of ordered pairs $\{(a, b) \mid a \in A \text{ and } f(a) = b\}$.



Graph of $f(n) = 2n + 1$
from $\mathbb{Z}$ to $\mathbb{Z}$



Graph of $f(x) = x^2$
from $\mathbb{Z}$ to $\mathbb{Z}$

Sequences

- Sequences are ordered lists of elements.
 - 1, 2, 3, 5, 8
 - 1, 3, 9, 27, 81,
- Sequences arise throughout mathematics, computer science, and in many other disciplines, ranging from botany to music.
- We will introduce the terminology to represent sequences and sums of the terms in the sequences.

Sequences

Definition: A *sequence* is a function from a subset of the integers (usually either the set $\{0, 1, 2, 3, 4, \dots\}$ or $\{1, 2, 3, 4, \dots\}$) to a set S .

- The notation a_n is used to denote the image of the integer n . We can think of a_n as the equivalent of $f(n)$ where f is a function from $\{0, 1, 2, \dots\}$ to S . We call a_n a *term* of the sequence.

Sequences

Example: Consider the sequence $\{a_n\}$ where

$$a_n = \frac{1}{n} \quad \{a_n\} = \{a_1, a_2, a_3, \dots\}$$

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4} \dots$$

Geometric Progression

Definition: A *geometric progression* is a sequence of the form: $a, ar, ar^2, \dots, ar^n, \dots$

where the *initial term* a and the *common ratio* r are real numbers.

Examples:

1. Let $a = 1$ and $r = -1$. Then:

$$\{b_n\} = \{b_0, b_1, b_2, b_3, b_4, \dots\} = \{1, -1, 1, -1, 1, \dots\}$$

2. Let $a = 2$ and $r = 5$. Then:

$$\{c_n\} = \{c_0, c_1, c_2, c_3, c_4, \dots\} = \{2, 10, 50, 250, 1250, \dots\}$$

3. Let $a = 6$ and $r = 1/3$. Then:

$$\{d_n\} = \{d_0, d_1, d_2, d_3, d_4, \dots\} = \{6, 2, \frac{2}{3}, \frac{2}{9}, \frac{2}{27}, \dots\}$$

Arithmetic Progression

Definition: A *arithmetic progression* is a sequence of the form: $a, a + d, a + 2d, \dots, a + nd, \dots$

where the *initial term* a and the *common difference* d are real numbers.

Examples:

1. Let $a = -1$ and $d = 4$:

$$\{s_n\} = \{s_0, s_1, s_2, s_3, s_4, \dots\} = \{-1, 3, 7, 11, 15, \dots\}$$

2. Let $a = 7$ and $d = -3$:

$$\{t_n\} = \{t_0, t_1, t_2, t_3, t_4, \dots\} = \{7, 4, 1, -2, -5, \dots\}$$

3. Let $a = 1$ and $d = 2$:

$$\{u_n\} = \{u_0, u_1, u_2, u_3, u_4, \dots\} = \{1, 3, 5, 7, 9, \dots\}$$

Useful Sequences

TABLE 1 Some Useful Sequences.

<i>n</i> th Term	First 10 Terms
n^2	1, 4, 9, 16, 25, 36, 49, 64, 81, 100, ...
n^3	1, 8, 27, 64, 125, 216, 343, 512, 729, 1000, ...
n^4	1, 16, 81, 256, 625, 1296, 2401, 4096, 6561, 10000, ...
2^n	2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, ...
3^n	3, 9, 27, 81, 243, 729, 2187, 6561, 19683, 59049, ...
$n!$	1, 2, 6, 24, 120, 720, 5040, 40320, 362880, 3628800, ...
f_n	1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, ...

Summations

- Sum of the terms $a_m, a_{m+1}, \dots, a_n$ from the sequence $\{a_n\}$
- The notation:

$$\sum_{j=m}^n a_j \quad \sum_{j=m}^n a_j \quad \sum_{m \leq j \leq n} a_j$$

represents

$$a_m + a_{m+1} + \dots + a_n$$

- The variable j is called the *index of summation*. It runs through all the integers starting with its *lower limit* m and ending with its *upper limit* n .

Summations

- More generally for a set S :

$$\sum_{j \in S} a_j$$

- **Examples:**

$$r^0 + r^1 + r^2 + r^3 + \cdots + r^n = \sum_0^n r^j$$

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots = \sum_1^{\infty} \frac{1}{i}$$

$$\text{If } S = \{2, 5, 7, 10\} \text{ then } \sum_{j \in S} a_j = a_2 + a_5 + a_7 + a_{10}$$

Prove that the sum of first n terms of an arithmetic progression is given by:

$$S_n = n/2 [2a + (n-1)d]$$

As we know that the sum of first n terms of an arithmetic series is given as

$$S_n = a + (a + d) + (a + 2d) + \dots + [a + (n-2)d] + [a + (n-1)d]$$

Let us write the series down again, but this time we shall write it down with the terms in reverse order. We get

$$S_n = [a + (n-1)d] + [a + (n-2)d] + \dots + (a + 2d) + (a + d) + a .$$

We are now going to add these two series together. On the left-hand side, we just get $2S_n$. But on the right-hand side, we are going to add the terms in the two series so that each term in the first series will be added to the term vertically below it in the second series. We get

$$2S_n = [2a + (n-1)d] + [2a + (n-1)d] + \dots + [2a + (n-1)d] + [2a + (n-1)d]$$

and on the right-hand side there are n copies of $[2a + (n-1)d]$ so we get

$$2S_n = n [2a + (n-1)d]$$

But of course we want S_n rather than $2S_n$, and so we divide by 2 to get

$$S_n = n/2 [2a + (n-1)d]$$

Example:

Find the sum of the first 50 terms of the sequence

1, 3, 5, 7, 9,

Solution:

This is an arithmetic progression, and we can write down

$a = 1$, $d = 2$, $n = 50$.

We now use the formula, so that

$$S_n = n/2 (2a + (n - 1)d)$$

$$S_{50} = 1/2 \times 50 \times (2 \times 1 + (50 - 1) \times 2)$$

$$= 25 \times (2 + 49 \times 2)$$

$$= 25 \times (2 + 98)$$

$$= 2500 .$$

The sum of the first n terms in a geometric sequence is given by

$$S_n = \frac{a_1(1 - r^n)}{1 - r}, r \neq 1$$

Proof: As we know that the sum of first n terms of a geometric series is given as

$$S_n = a + ar + ar^2 + ar^3 + \dots + ar^{n-1},$$

Lets multiply this equation by r and then subtract:

$$r S_n = ar + ar^2 + ar^3 + \dots + ar^{n-1} + ar^n$$

$$S_n - rS_n = a - ar^n$$

$$S_n(1 - r) = a(1 - r^n).$$

Now divide by $1 - r$ (as long as $r \neq 1$) to give

$$S_n = a(1 - r^n) / (1 - r)$$

Find the sum of the geometric series

$$2 + 6 + 18 + 54 + \dots$$

where there are 6 terms in the series.

Solution

For this series, we have $a = 2$, $r = 3$ and $n = 6$. So

$$S_n = a(1 - r^n) / (1 - r)$$

$$S_6 = 2(1 - 3^6) / (1 - 3)$$

$$= 2(1 - 729) / (-2)$$

$$= -(-728)$$

$$= 728$$

Some Useful Summation Formulae

TABLE 2 Some Useful Summation Formulae.

<i>Sum</i>	<i>Closed Form</i>
$\sum_{k=0}^n ar^k \ (r \neq 0)$	$\frac{ar^{n+1} - a}{r - 1}, r \neq 1$
$\sum_{k=1}^n k$	$\frac{n(n+1)}{2}$
$\sum_{k=1}^n k^2$	$\frac{n(n+1)(2n+1)}{6}$
$\sum_{k=1}^n k^3$	$\frac{n^2(n+1)^2}{4}$
$\sum_{k=0}^{\infty} x^k, x < 1$	$\frac{1}{1-x}$
$\sum_{k=1}^{\infty} kx^{k-1}, x < 1$	$\frac{1}{(1-x)^2}$

Geometric Series: We just proved this.

Later we will prove some of these by induction.

Proof in text (requires calculus)

Cardinality

Definition: The *cardinality* of a set A is equal to the cardinality of a set B , denoted

$$|A| = |B|,$$

if and only if there is a one-to-one correspondence (*i.e.*, a bijection) from A to B .

- If there is a one-to-one function (*i.e.*, an injection) from A to B , the cardinality of A is less than or the same as the cardinality of B and we write $|A| \leq |B|$.
- When $|A| \leq |B|$ and A and B have different cardinality, we say that the cardinality of A is less than the cardinality of B and write $|A| < |B|$.

Matrix

Definition: A *matrix* is a rectangular array of numbers. A matrix with m rows and n columns is called an $m \times n$ matrix.

- The plural of matrix is *matrices*.
- A matrix with the same number of rows as columns is called *square*.
- Two matrices are *equal* if they have the same number of rows and the same number of columns and the corresponding entries in every position are equal.

3×2 matrix

$$\begin{bmatrix} 1 & 1 \\ 0 & 2 \\ 1 & 3 \end{bmatrix}$$

Matrix Arithmetic: Addition

Defintion: Let $\mathbf{A} = [a_{ij}]$ and $\mathbf{B} = [b_{ij}]$ be $m \times n$ matrices. The sum of $\mathbf{A}$ and $\mathbf{B}$, denoted by $\mathbf{A} + \mathbf{B}$, is the $m \times n$ matrix that has $a_{ij} + b_{ij}$ as its (i,j) th element. In other words, $\mathbf{A} + \mathbf{B} = [a_{ij} + b_{ij}]$.

Example:

$$\begin{bmatrix} 1 & 0 & -1 \\ 2 & 2 & -3 \\ 3 & 4 & 0 \end{bmatrix} + \begin{bmatrix} 3 & 4 & -1 \\ 1 & -3 & 0 \\ -1 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 4 & 4 & -2 \\ 3 & -1 & -3 \\ 2 & 5 & 2 \end{bmatrix}$$

Note that matrices of different sizes can not be added.

Matrix Multiplication

Definition: Let $\mathbf{A}$ be an $m \times k$ matrix and $\mathbf{B}$ be a $k \times n$ matrix. The *product* of $\mathbf{A}$ and $\mathbf{B}$, denoted by $\mathbf{AB}$, is the $m \times n$ matrix that has its (i,j) th element equal to the sum of the products of the corresponding elements from the i th row of $\mathbf{A}$ and the j th column of $\mathbf{B}$. In other words, if $\mathbf{AB} = [c_{ij}]$ then $c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{ik}b_{kj}$.

Example:

$$\begin{bmatrix} 1 & 0 & 4 \\ 2 & 1 & 1 \\ 3 & 1 & 0 \\ 0 & 2 & 2 \end{bmatrix} \begin{bmatrix} 2 & 4 \\ 1 & 1 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} 14 & 4 \\ 8 & 9 \\ 7 & 13 \\ 8 & 2 \end{bmatrix}$$

The product of two matrices is undefined when the number of columns in the first matrix is not the same as the number of rows in the second.

Illustration of Matrix Multiplication

- The Product of $\mathbf{A} = [a_{ij}]$ and $\mathbf{B} = [b_{ij}]$

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1k} \\ a_{21} & a_{22} & \dots & a_{2k} \\ \vdots & \vdots & & \vdots \\ \vdots & \vdots & & \vdots \\ \textcolor{red}{a_{i1}} & \textcolor{red}{a_{i2}} & \dots & \textcolor{red}{a_{ik}} \\ \vdots & \vdots & & \vdots \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mk} \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} b_{11} & b_{12} & \dots & \textcolor{red}{b_{1j}} & \dots & b_{1n} \\ b_{21} & b_{22} & \dots & \textcolor{red}{b_{2j}} & \dots & b_{2n} \\ \vdots & \vdots & & \vdots & & \vdots \\ \vdots & \vdots & & \vdots & & \vdots \\ b_{k1} & b_{k2} & \dots & \textcolor{red}{b_{kj}} & \dots & b_{kn} \end{bmatrix}$$

$$\mathbf{AB} = \begin{bmatrix} c_{11} & c_{12} & \dots & c_{1n} \\ c_{21} & c_{22} & \dots & c_{2n} \\ \vdots & \vdots & & \vdots \\ \vdots & \vdots & \textcolor{red}{c_{ij}} & \vdots \\ \vdots & \vdots & & \vdots \\ c_{m1} & c_{m2} & \dots & c_{mn} \end{bmatrix}$$

$$\textcolor{red}{c_{ij}} = \textcolor{red}{a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{ik}b_{kj}}$$

Matrix Multiplication is not Commutative

Example: Let

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}$$

$$\mathbf{B} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

Does $\mathbf{AB} = \mathbf{BA}$?

Solution:

$$\mathbf{AB} = \begin{bmatrix} 2 & 2 \\ 5 & 3 \end{bmatrix}$$

$$\mathbf{BA} = \begin{bmatrix} 4 & 3 \\ 3 & 2 \end{bmatrix}$$

$$\mathbf{AB} \neq \mathbf{BA}$$

Identity Matrix and Powers of Matrices

Definition: The *identity matrix* of order n is the $m \times n$ matrix $\mathbf{I}_n = [\delta_{ij}]$, where $\delta_{ij} = 1$ if $i = j$ and $\delta_{ij} = 0$ if $i \neq j$.

$$\mathbf{I}_n = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \cdot & \cdot & & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & & \cdot \\ 0 & 0 & \dots & 1 \end{bmatrix}$$

$$\mathbf{A}\mathbf{I}_n = \mathbf{I}_m\mathbf{A} = \mathbf{A}$$

when $\mathbf{A}$ is an $m \times n$ matrix

Powers of square matrices can be defined. When $\mathbf{A}$ is an $n \times n$ matrix, we have:

$$\mathbf{A}^0 = \mathbf{I}_n \quad \mathbf{A}^r = \underbrace{\mathbf{A}\mathbf{A}\mathbf{A}\dots\mathbf{A}}_{r \text{ times}}$$

Transposes of Matrices

Definition: Let $\mathbf{A} = [a_{ij}]$ be an $m \times n$ matrix. The *transpose* of $\mathbf{A}$, denoted by $\mathbf{A}^t$, is the $n \times m$ matrix obtained by interchanging the rows and columns of $\mathbf{A}$.

If $\mathbf{A}^t = [b_{ij}]$, then $b_{ij} = a_{ji}$ for $i = 1, 2, \dots, n$
and $j = 1, 2, \dots, m$.

The transpose of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ is the matrix $\begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$.

Transposes of Matrices

Definition: A square matrix $\mathbf{A}$ is called symmetric if $\mathbf{A} = \mathbf{A}^t$. Thus $\mathbf{A} = [a_{ij}]$ is symmetric if $a_{ij} = a_{ji}$ for i and j with $1 \leq i \leq n$ and $1 \leq j \leq n$.

The matrix $\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ is square.

Square matrices do not change when their rows and columns are interchanged.

Zero-One Matrices

Definition: A matrix all of whose entries are either 0 or 1 is called a *zero-one matrix*. (These will be used in Chapters 9 and 10.)

Algorithms operating on discrete structures represented by zero-one matrices are based on Boolean arithmetic defined by the following Boolean operations:

$$b_1 \wedge b_2 = \begin{cases} 1 & \text{if } b_1 = b_2 = 1 \\ 0 & \text{otherwise} \end{cases} \quad b_1 \vee b_2 = \begin{cases} 1 & \text{if } b_1 = 1 \text{ or } b_2 = 1 \\ 0 & \text{otherwise} \end{cases}$$

Zero-One Matrices

Definition: Let $\mathbf{A} = [a_{ij}]$ and $\mathbf{B} = [b_{ij}]$ be an $m \times n$ zero-one matrices.

- The *join* of $\mathbf{A}$ and $\mathbf{B}$ is the zero-one matrix with (i,j) th entry $a_{ij} \vee b_{ij}$. The *join* of $\mathbf{A}$ and $\mathbf{B}$ is denoted by $\mathbf{A} \vee \mathbf{B}$.
- The *meet* of $\mathbf{A}$ and $\mathbf{B}$ is the zero-one matrix with (i,j) th entry $a_{ij} \wedge b_{ij}$. The *meet* of $\mathbf{A}$ and $\mathbf{B}$ is denoted by $\mathbf{A} \wedge \mathbf{B}$.

Joins and Meets of Zero-One Matrices

Example: Find the join and meet of the zero-one matrices

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}.$$

Solution: The join of $\mathbf{A}$ and $\mathbf{B}$ is

$$\mathbf{A} \vee \mathbf{B} = \begin{bmatrix} 1 \vee 0 & 0 \vee 1 & 1 \vee 0 \\ 0 \vee 1 & 1 \vee 1 & 0 \vee 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

The meet of $\mathbf{A}$ and $\mathbf{B}$ is

$$\mathbf{A} \wedge \mathbf{B} = \begin{bmatrix} 1 \wedge 0 & 0 \wedge 1 & 1 \wedge 0 \\ 0 \wedge 1 & 1 \wedge 1 & 0 \wedge 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

Boolean Product of Zero-One Matrices

Definition: Let $\mathbf{A} = [a_{ij}]$ be an $m \times k$ zero-one matrix and $\mathbf{B} = [b_{ij}]$ be a $k \times n$ zero-one matrix. The *Boolean product* of $\mathbf{A}$ and $\mathbf{B}$, denoted by $\mathbf{A} \odot \mathbf{B}$, is the $m \times n$ zero-one matrix with (i,j) th entry

$$c_{ij} = (a_{i1} \wedge b_{1j}) \vee (a_{i2} \wedge b_{2j}) \vee \dots \vee (a_{ik} \wedge b_{kj}).$$

Example: Find the Boolean product of $\mathbf{A}$ and $\mathbf{B}$, where

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix},$$

$$\mathbf{B} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

Continued on next slide →

Boolean Product of Zero-One Matrices

Solution: The Boolean product $\mathbf{A} \odot \mathbf{B}$ is given by

$$\begin{aligned}\mathbf{A} \odot \mathbf{B} &= \begin{bmatrix} (1 \wedge 1) \vee (0 \wedge 0) & (1 \wedge 1) \vee (0 \wedge 1) & (1 \wedge 0) \vee (0 \wedge 1) \\ (0 \wedge 1) \vee (1 \wedge 0) & (0 \wedge 1) \vee (1 \wedge 1) & (0 \wedge 0) \vee (1 \wedge 1) \\ (1 \wedge 1) \vee (0 \wedge 0) & (1 \wedge 1) \vee (0 \wedge 1) & (1 \wedge 0) \vee (0 \wedge 1) \end{bmatrix} \\ &= \begin{bmatrix} 1 \vee 0 & 1 \vee 0 & 0 \vee 0 \\ 0 \vee 0 & 0 \vee 1 & 0 \vee 1 \\ 1 \vee 0 & 1 \vee 0 & 0 \vee 0 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}.\end{aligned}$$

Boolean Powers of Zero-One Matrices

Definition: Let $\mathbf{A}$ be a square zero-one matrix and let r be a positive integer. The r th Boolean power of $\mathbf{A}$ is the Boolean product of r factors of $\mathbf{A}$, denoted by $\mathbf{A}^{[r]}$. Hence,

$$\mathbf{A}^{[r]} = \underbrace{\mathbf{A} \odot \mathbf{A} \odot \dots \odot \mathbf{A}}_{r \text{ times}}.$$

We define $\mathbf{A}^{[r]}$ to be $\mathbf{I}_n$.

(The Boolean product is well defined because the Boolean product of matrices is associative.)

Boolean Powers of Zero-One Matrices

Example: Let $\mathbf{A} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix}$.

Find $\mathbf{A}^n$ for all positive integers n .

Solution:

$$\mathbf{A}^{[2]} = \mathbf{A} \odot \mathbf{A} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix} \quad \mathbf{A}^{[3]} = \mathbf{A}^{[2]} \odot \mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\mathbf{A}^{[4]} = \mathbf{A}^{[3]} \odot \mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\mathbf{A}^{[5]} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \quad \mathbf{A}^{[n]} = \mathbf{A}^{[5]} \quad \text{for all positive integers } n \text{ with } n \geq 5.$$